

## ENFORCER

boat: arrow 2

### A A fine becomes a price

*Breaks where the rule meets behavior*

#### MECHANISM

##### Fine

a charge for a violation

#### FAILURE

##### **Paid, not obeyed**

agents treat the fine as a fee for the act

#### MEASURE

contribution stability

#### CHECK

**Does cooperation hold without the fine?**

yes: keep and retest · no: add safeguards

#### SAFEGUARDS

- ✓ state the norm, not only the price
- ✓ graduated sanctions

MOST EXPOSED **incentive**

↻ revise, then test again

## OBSERVER

boat: arrow 4

### B A score becomes the target

*Breaks where behavior becomes a signal*

#### MECHANISM

##### Reputation score

past conduct sets trust

#### FAILURE

##### **Score is gamed**

reputation farming; shallow cooperation

#### MEASURE

gameability; false negatives

#### CHECK

**Does the score match real conduct?**

yes: keep and retest · no: add safeguards

#### SAFEGUARDS

- ✓ tamper-evident records
- ✓ peer review
- ✓ audit messages, not only scores

MOST EXPOSED **social** **epistemic**

↻ revise, then test again

## ARBITER

boat: the feedback loop

### C A role becomes an authority

*Breaks where the institution feeds back on itself*

#### MECHANISM

##### Monitor or judge

a role that can sanction

#### FAILURE

##### **Over-detects or is captured**

rewarded for finding violations

#### MEASURE

false punishment; capture; override

#### CHECK

**Are rulings accurate and reversible?**

yes: keep and retest · no: add safeguards

#### SAFEGUARDS

- ✓ checks on sanctioners
- ✓ appeals
- ✓ elected or rotating roles
- ✓ a human stop

MOST EXPOSED **epistemic** **constraint**

↻ revise, then test again